You were redirected here from etc:opencv:start.

OpenCV Installation

OpenCV docs : https://docs.opencv.org/master/ (https://docs.opencv.org/master/)

Update

```
$ sudo apt update
$ sudo apt -y upgrade
$ sudo apt -y autoremove
```

Required packages

The required packages are build-essential, cmake, git, python, libgtk2.0-dev, pkg-config, libavcodec-dev, libavformat-dev, and libswscale-dev.

The others are optional.

Python

```
$ sudo apt install -y python-dev python3-dev python3-pip
$ python3 -m pip install -U pip
$ python3 -m pip install --user numpy
```

GCC, CMake, git

```
$ sudo apt install -y build-essential cmake git
```

GUI, GL

```
$ sudo apt install -y libgtk2.0-dev libgtk-3-dev libqt4-dev mesa-ut
ils libgl1-mesa-dri libqt4-opengl-dev
```

config

```
$ sudo apt install -y pkg-config
```

FFmpeg libraries

```
$ sudo apt install -y libavcodec-dev libavformat-dev libswscale-dev
```

Optional packages

parallelism library

```
$ sudo apt install -y libtbb2 libtbb-dev
```

Image codec

```
$ sudo apt install -y libjasper-dev libjpeg-dev libtiff5-dev
$ sudo apt install -y libpng12-dev || sudo apt install -y libpng-de
v
```

When the installation of libjasper-dev fails, download source from https://github.com/mdadams/jasper (https://github.com/mdadams/jasper) and build according to the following command.

If you skip installing libjasper-dev, it will be built during building OpenCV.

```
$ sudo apt install -y freeglut3-dev
$ git clone https://github.com/mdadams/jasper.git
$ cd jasper/build
$ cmake ..
$ sudo make install
```

Video codec

```
$ sudo apt install -y libxvidcore-dev libx264-dev libxine2-dev libv
4l-dev v4l-utils
```

Streming

```
$ sudo apt install -y libgstreamer1.0-dev libgstreamer-plugins-base
1.0-dev
```

IEEE 1394 digital camera

```
$ sudo apt install -y libdc1394-22-dev
```

Mathmatic

```
$ sudo apt install -y gfortran libatlas-base-dev libeigen3-dev libh
df5-serial-dev liblapack-dev
```

Building OpenCV from Source Using CMake

Getting OpenCV Source Code

```
$ cd $HOME
$ git clone https://github.com/opencv/opencv_contrib.git
$ cd opencv_contrib
$ git checkout tags/4.1.0 -b v4.1.0
```

```
$ cd $HOME
$ git clone https://github.com/opencv/opencv.git
$ cd opencv
$ git checkout tags/4.1.0 -b v4.1.0
```

CMake

If you want to optimize OpenCV using **OpenCL**, See the CMake section of this link before runnig CMake.

```
$ cd ~/opencv
$ mkdir build
$ cd build
$ cmake ..\
    -DCMAKE_BUILD_TYPE=Release \
    -DOPENCV_GENERATE_PKGCONFIG=ON \
    -DBUILD_PERF_TESTS=OFF \
    -DOPENCV_EXTRA_MODULES_PATH=$HOME/opencv_contrib/modules
```

Checking memory size and swapon

Make sure the memory size(Mem + Swap) is at least 3 GB () before building.

```
$ free -ht
              total        used        free      shared  buff/cache
available
Mem:           1.8G        635M        219M         29M        1.0G
1.2G
Swap:            0B          0B          0B
Total:         1.8G        635M        219M
```

If you don't have enough memory like above, increase the swap. If you want to use `-j[N-jobs]` option with `make`, need more than 3 GB () memory size.

Set as many sizes of swapfile as it needs to compile. And register it as swap.

```
$ sudo fallocate -l 1.2G /swapfile
```

```
$ sudo chmod 600 /swapfile
$ sudo mkswap /swapfile
$ sudo swapon /swapfile
```

build

```
$ make
```

If memory and disk size are enough, run `make -j $(expr $(expr $(nproc) \* 6) \/ 5)` instead of `make` for quick compilation.

Core	Disk	RAM	Swap	Command
4 core(C1+)	16 GB ()	0.8 GB ()	2.4 GB ()	make -j4
4 core(C2)	16 GB ()	1.7 GB ()	1.6 GB ()	make -j4
6 core(N2)	16 GB ()	1.8 GB ()	1.0 GB ()	make
6 core(N2)	16 GB ()	1.8 GB ()	1.5 GB ()	make -j7
6 core(N2)	16 GB ()	3.6 GB ()	0 GB ()	make -j7
8 core(XU4)	16 GB ()	1.9 GB ()	1.4 GB ()	make -j9

After compilation is complete, release the swap and delete the swapfile.

```
$ sudo swapoff /swapfile
$ sudo rm /swapfile
```

To install libraries, execute the following command from 'build' directory

```
$ sudo make install
$ sudo ldconfig
```

C++ Compilation

```
$ g++ [input.cpp] -o [output] $(pkg-config opencv4 --libs --cflags)
```

Python

virtual environment

Create a virtual environment.

```
$ python3 -m venv [venv_path]
```

Check user's cv2 package path.

```
$ sudo find / -name cv2 -type d
/usr/local/lib/python3.6/dist-packages/cv2
```

Copy python cv2 package and paste it in the virtual environment package path.

```
$ sudo cp -r /usr/local/lib/python3.6/dist-packages/cv2 [venv_path]/lib/python3.6/site-packages/
```

Activate the virtual environment.

```
$ source [venv_path]/bin/activate
```

When using OpenCV-python, numpy and matplotlib are used together. (numpy is required, matplotlib is optional.)

```
$ python3 -m pip install -U pip &&\
  python3 -m pip install numpy matplotlib
```

To test if OpenCV module is imported properly, execute the command below.

```
$ python3 -c "import cv2 as cv"
```

etc/opencv/installation.txt Last modified: 2024/05/17 16:44 by tobetter

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